

Prob 1

$$2y'' + y' - y = 3u(t-2), \quad y(0) = y'(0) = 0$$

$$2L(y'') + L(y') - L(y) = 3L(u(t-2))$$

$$2(s^2 F(s) - s f(0) - f'(0)) + (s F(s) - f(0)) - F(s) = \frac{3}{s} e^{-2s}$$

$$2s^2 F(s) + s F(s) - F(s) = \frac{3}{s} e^{-2s}$$

$$F(s)(2s^2 + s - 1) = \frac{3}{s} e^{-2s}$$

$$F(s) = \frac{3e^{-2s}}{s(2s^2 + s - 1)} \Rightarrow 3 = A(2s^2 + s - 1) + (Bs + C)s = 2As^2 + As - A + Bs^2 + Cs$$

$$\Rightarrow 2A + B = 0 \Rightarrow B = -2A = -6$$

$$F(s) = e^{-2s} \left(-\frac{3}{s} + \frac{6s}{(2s-1)(s+1)} + \frac{3}{(2s-1)(s+1)} \right)$$

$$A + C = 0 \Rightarrow C = -A = 3$$

$$-A = 3 \Rightarrow A = -3$$

$$6s = A(s+1) + B(2s-1) \Rightarrow 6s = s(A+2B) + A-B \Rightarrow A+2B=6$$

$$A-B=0 \Rightarrow A=B=2$$

$$3 = A(s+1) + B(2s-1) \Rightarrow 3 = s(A+2B) + A-B \Rightarrow A+2B=0$$

$$A-B=3 \Rightarrow B=1 \wedge A=2$$

$$F(s) = e^{-2s} \left(-\frac{3}{s} + \frac{4}{2s-1} + \frac{1}{s+1} \right)$$

$$f(t) = F^{-1}(s) = u(t-2) \left(-3 + 2e^{\frac{1}{2}t-2} + e^{-(t-2)} \right)$$

Prob 2

$$g(t) \Rightarrow G(s) = \frac{1}{s-2}$$

$$y'(t) + 5 \int_0^t e^{2\tau} y(t-\tau) d\tau = e^{2t}, \quad y(0) = 0$$

$$L(y'(t)) = sF(s) + f(0)$$

$$L\left(\int_0^t F(t-\tau) g(\tau) d\tau\right) = F(s) G(s)$$

$$sF(s) + f(0) + 5F(s)G(s) = \frac{1}{s-2}$$

$$sF(s) + 5F(s)\frac{1}{s-2} = \frac{1}{s-2}$$

$$s(s-2)F(s) + 5F(s) = 1$$

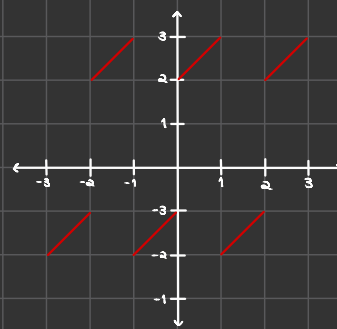
$$F(s)(s^2 - 2s + 5) = 1$$

$$F(s) = \frac{1}{s^2 - 2s + 5} = \frac{1}{(s-1)^2 + 4} = \frac{1}{(s-1)^2 + 2^2} = \frac{1}{2} \left(\frac{2}{(s-1)^2 + 2^2} \right) \Rightarrow \frac{1}{2} \sin(2t) e^t$$

Prob 3

$$f(x) = 2 + x, \quad 0 \leq x \leq 1$$

We know that g is odd. That gives $a_0 = a_n = 0$.



odd · odd = even

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx && \text{Feil greenser } \rightarrow 0 \rightarrow 1 \\
 &= \frac{2}{3} \int_0^3 (2+x) \sin\left(\frac{n\pi x}{3}\right) dx \\
 &= \frac{4}{3} \int_0^3 \sin\left(\frac{n\pi x}{3}\right) dx + \frac{2}{3} \int_0^3 x \sin\left(\frac{n\pi x}{3}\right) dx \\
 &= -\frac{4}{3} \left[\frac{3 \cos\left(\frac{n\pi x}{3}\right)}{n\pi} \right]_0^3 + \frac{2}{3} \left(-x \cdot \frac{3 \cos\left(\frac{n\pi x}{3}\right)}{n\pi} + \int_0^3 \frac{3 \cos\left(\frac{n\pi x}{3}\right)}{n\pi} dx \right) \\
 &= -\frac{4}{n\pi} (\cos(n\pi) - 1) + \frac{2}{n\pi} \left[-x \cos\left(\frac{n\pi x}{3}\right) + \int \cos\left(\frac{n\pi x}{3}\right) dx \right]_0^3 \\
 &= -\frac{4}{n\pi} (\cos(n\pi) - 1) + \frac{2}{n\pi} \left[-x \cos\left(\frac{n\pi x}{3}\right) + \frac{3 \sin\left(\frac{n\pi x}{3}\right)}{n\pi} \right]_0^3 \\
 &= \frac{2}{n\pi} (2 - 2 \cos(n\pi) - 3 \cos(n\pi) + \frac{3 \sin(n\pi)}{n\pi}) \\
 &= \frac{2}{n\pi} (2 - 5 \cos(n\pi) + \frac{3 \sin(n\pi)}{n\pi}) \\
 &= \frac{2}{\pi n} (2 - 2 \cos(n\pi) - \cos(n\pi) + \frac{\sin(n\pi)}{\pi n} \Big|_{0 \rightarrow 0}^{1 \rightarrow 0}) \\
 &= \frac{2}{\pi n} (2 - 3 \cos(n\pi)) \\
 &= \underline{\underline{\frac{4 - 6(-1)^n}{\pi n}}}
 \end{aligned}$$

Prob 4

$$u_t = 4u_{xx}, \quad c=2, \quad L=1, \quad -1 < x < 1, \quad u(-1, t) = u(1, t) = 0$$

$$u(x, t) = F(x) G(t)$$

$$F(x) G'(t) = 4 F''(x) G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{1}{4} \frac{G'(t)}{G(t)} = k$$

$$\frac{F''(x)}{F(x)} = k, \quad \frac{1}{4} \frac{G'(t)}{G(t)} = k, \quad k = -p^2$$

$$F''(x) + p^2 F(x) = 0 \quad \wedge \quad G'(t) + 4p^2 G(t) = 0$$

$$F(x) = A \cos(px) + B \sin(px)$$

$$F(-1) = A \cos(-p) + B \sin(-p) = A \cos(p) - B \sin(p) = 0 \Rightarrow A = 0$$

$$F(1) = A \cos(p) + B \sin(p) = 0$$

$$F(1) + F(-1) = 2 \cos(p) = 0 \Rightarrow p = \left(n + \frac{1}{2}\right) \pi \Rightarrow k = -p^2 = -\left(n + \frac{1}{2}\right)^2 \pi^2$$

$$F(1) - F(-1) = 2 B \sin(p) = 0 \Rightarrow p = n\pi \Rightarrow k = -p^2 = -n^2 \pi^2$$

$$F(x) = \cos\left(\left(n + \frac{1}{2}\right) \pi x\right)$$

✓

$$F(x) = \sin(n\pi x)$$

$$G = G_0 e^{c_k t} = G_0 e^{-4\left(n + \frac{1}{2}\right)^2 \pi^2 t}$$

✓

$$G(x) = G_0 e^{c_k t} = G_0 e^{-4n^2 \pi^2 t}$$

Prob 6

$$u'(x) + O(h^2) = \frac{-3u(x) + 4u(x+h) - u(x+2h)}{2h}$$

$$u(x+h) = u(x) + u'(x)h + O(h^2) \Rightarrow 4u(x+h) = 4u(x) + 4u'(x)h + 4O(h^2)$$

$$u(x+2h) = u(x) + 2u'(x)h + O(h^2)$$

$$4u(x+h) - u(x+2h) = 4u(x) + 4u'(x)h + 4O(h^2) - (u(x) + 2u'(x)h + O(h^2))$$

$$4u(x+h) - u(x+2h) = 3u(x) + 2u'(x)h + 5O(h^2)$$

$$u'(x) + O(h^2) = \frac{4u(x+h) - u(x+2h) - 3u(x)}{2h}$$

$$O(h^2) = O(\epsilon/h) \Rightarrow h^3 = O(\epsilon) \Rightarrow h = O(\sqrt[3]{\epsilon}) \Rightarrow p = \frac{1}{3}$$

Prob 7

$$x = \sqrt{x} + 2 = g(x)$$

a) Need to show that:

1) $g(x)$ is continuously differentiable on $x \in [1, 9]$ and that $g(x) \in [1, 9] \quad \forall x \in [1, 9]$

$$g(x) = \sqrt{x} + 2 \Rightarrow g'(x) = \frac{1}{2\sqrt{x}}, \quad g(1) = 3, \quad g(9) = 5 \Rightarrow 1 \leq g(x) \leq 9 \quad \forall x \in [1, 9]$$

2) $g'(x)$ exists $\forall x \in [1, 9]$ and that there is a pos. const. L such that $g'(x) \leq L < 1 \quad \forall x \in [1, 9]$

$$g'(1) = \frac{1}{2}, \quad g'(9) = \frac{1}{6} \Rightarrow g'(x) \leq L < 1 \quad \forall x \in [1, 9], \quad L = \frac{1}{2}$$

b) $|x_{k+1} - r| = 2^{-10}, \quad L = \frac{1}{2}, \quad x_0 = 2, \quad g(x_0) = \sqrt{2} + 2, \quad g(x_0) - x_0 = \sqrt{2}$

$$|x_{k+1} - r| \leq \frac{L^{k+1}}{L} |g(x_0) - x_0|$$

$$2^{-10} = \frac{\frac{1}{2}^{k+1}}{\frac{1}{2}} \sqrt{2} \Rightarrow \frac{1}{2^{10}} = \frac{1}{2^k} \sqrt{2} \Rightarrow \frac{1}{2^{10}\sqrt{2}} = \frac{1}{2^k} \Rightarrow 2^{\frac{21}{2}} = 2^k \Rightarrow \frac{21}{2} \log(2) = k \log(2) \Rightarrow k = \frac{21}{2}$$

Prob 8

0	0	
1/2	1/2	0
1	1/2	1/2
	1/6	1/6

$$\sum b_i = \frac{1}{6} + \frac{4}{6} + \frac{1}{6} = 1 \Rightarrow p = 1 \text{ OK!}$$

$$\sum b_i c_i = \frac{1}{6} \cdot 0 + \frac{4}{6} \cdot \frac{1}{2} + \frac{1}{6} \cdot 1 = \frac{4}{12} + \frac{2}{12} = \frac{1}{2} \Rightarrow p = 2 \text{ OK!}$$

$$\sum b_i c_i^2 = \frac{1}{6} \cdot 0^2 + \frac{4}{6} \cdot \frac{1}{2}^2 + \frac{1}{6} \cdot 1^2 = \frac{4}{24} + \frac{4}{24} = \frac{1}{3} \Rightarrow p = 3$$

$$\sum \sum b_i a_{ij} c_j = 0 + \frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{2} + 0 = \frac{1}{24} \neq \frac{1}{6} \Rightarrow p \neq 3$$

Prob 9

	0	0	0
	1/2	1/2	0
s=3	3/4	0	3/4
		2/9	1/3
			4/9

$$k_1 = f(x_n, y_n) \quad k_2 = f(x_n + h, y_n + h k_1), \quad h_0 = 0,2 \quad y' = f(x_n, y_n)$$

$$k_i = f(x_n + c_i h, y_n + h \sum_{j=1}^s a_{ij} k_j)$$

$$k_1 = f(x_n + c_1 h, y_n + h \sum_{j=1}^s a_{1j} k_j) = f(x_n, y_n)$$

$$k_2 = f(x_n + c_2 h, y_n + h \sum_{j=1}^s a_{2j} k_j) = f(x_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$$

$$k_3 = f(x_n + c_3 h, y_n + h \sum_{j=1}^s a_{3j} k_j) = f(x_n + \frac{3h}{4}, y_n + \frac{3h}{4} k_2)$$

$$y_{n+1} = y_n + h \sum_{i=1}^s b_i k_i = y_n + h (\frac{2}{9} k_1 + \frac{1}{3} k_2 + \frac{4}{9} k_3)$$

$$k_1 = f(x_1, y_1) = f(0, 1) \Rightarrow y' = -y^2 = -1^2$$

$$k_2 = f(x_1 + \frac{h}{2}, y_1 + \frac{h}{2} k_1) = f(0 + \frac{0,2}{2}, 1 + \frac{0,2}{2} \cdot (-1)) = f(0,1, 0,9) \Rightarrow y' = -y^2 = -0,81 \quad \text{// } k_2$$

$$k_3 = f(x_n + \frac{3h}{4}, y_2 + \frac{3h}{4} k_2) = f(0,1 + \frac{3 \cdot 0,2}{4}, 0,9 + \frac{3 \cdot 0,2}{4} \cdot (-0,81)) = f(0,25, 0,8785)$$

$$0,9 - \frac{3}{4} \cdot 0,81 = 0,8785$$

$$\frac{3}{4} \cdot 0,81$$

Prob 10

$$u'' + x^2 u = \sin(\pi x), \quad 0 \leq x \leq 1, \quad u'(0) = 0, \quad u(1) = 0$$

$$u(x+h) = u(x) + u'(x)h + \frac{u''(x)h^2}{2}$$

$$u(x-h) = u(x) - u'(x)h + \frac{u''(x)h^2}{2}$$

$$u(x+h) + u(x-h) = u(x) + \cancel{u'(x)h} + \frac{u''(x)h^2}{2} + (u(x) - \cancel{u'(x)h} + \frac{u''(x)h^2}{2})$$

$$u''(x) = \frac{u(x+h) + u(x-h) - 2u(x)}{h^2} = \frac{u^{i+1} + u^{i-1} - 2u^i}{h^2}$$

$$u'' + x^2 u = \frac{u^{i+1} + u^{i-1} - 2u^i}{h^2} + x^2 u^i = \frac{u^{i+1}}{h^2} + \frac{u^{i-1}}{h^2} + u^i (x^2 - \frac{2}{h^2}) = f^i$$